



Broadband Driver Amplifier Family

Product Overview

MMIC Product Line

Features

- Bandwidth DC to 65 GHz
- Data rates from 39 to 44 Gb/s
- Implemented in GaAs PHEMT technology

Applications

- Fiber Optic Telecommunications systems (SONET/SDH OC-768/STM-256)
- Fiber Optic Test Equipment
- Generic broadband amplifier applications in test systems and military electronics

Product Summary

Model	Description	Application	Status
DA1	BW >65GHz V p-p >3.5V Gain 10 dB	Electro-Absorption Modulator Driver Trans-Impedance Amplifier General Purpose Gain Block	Released
DA2	BW >65GHz V p-p >6V Gain 10 dB	Polymer Modulator Driver Electro-Absorption Modulator Driver General Purpose Gain Block	Released
DA3	BW >65GHz V p-p >7.5V Gain 10 dB	Lithium Niobate Modulator Driver General Purpose PA	Released
DA4	DA1 with integrated LF-X bias network (patent pending)	Allows frequency extension to 30 kHz without inductor	Released

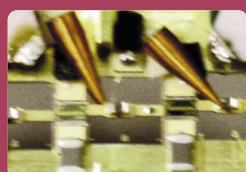
Advantages

- Fast rise/fall times (<8 ps)
- Low Jitter (<500 fs rms)
- Low Power Dissipation

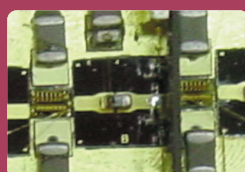
LF-X™—A Centellax Innovation

LF-X™

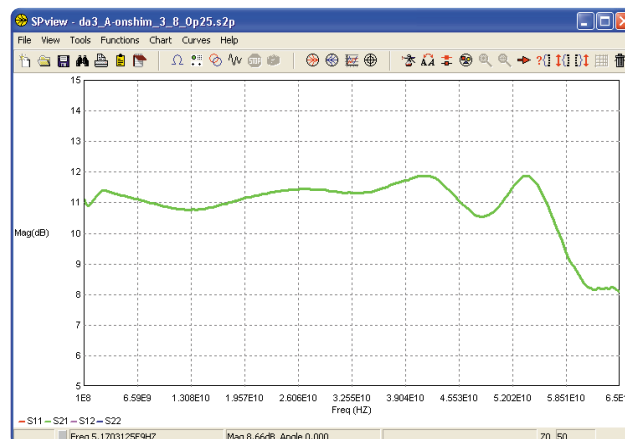
- Unique biasing without high Q coils
- Frequency extension to 30 kHz (Centellax patent pending)



to



The Centellax LF-X (patent pending) on-chip bias circuit eliminates the need for broadband external coils.



Broadband S-parameter data for DA3



Amplifier Module Product Line

Features

- Bandwidth DC to 65 GHz
- Data rates from 39 to 44 Gb/s
- Self-contained connectorized packages
- EA and LN Applications

Applications

- Fiber Optic Telecommunications systems (SONET/SDH OC-768/STM-256)
- Fiber Optic Test Equipment
- Generic broadband amplifier applications in test systems and military electronics

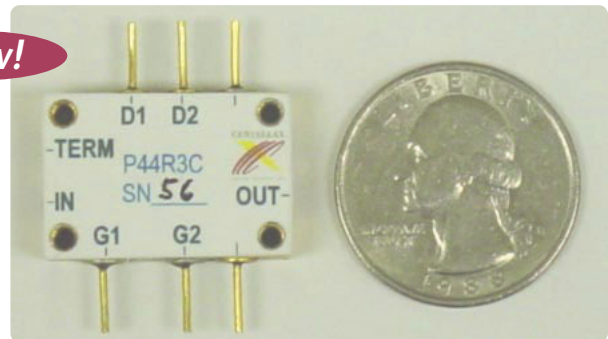
Advantages

- Fast rise/fall times (<8 ps)
- Low Jitter (<500 fs rms)
- Low Power Dissipation

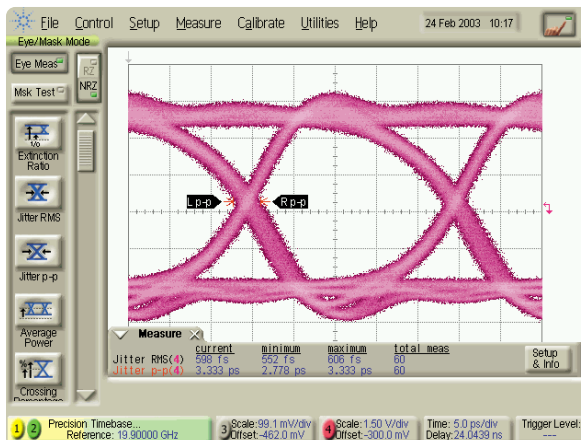
Product Summary

Model	Description	Application	Status
P44R3C	BW >65GHz, V p-p >3.0V 20 dB gain, Pdc 1.2W, <500 fs rms jitter, GPPO connectors	Short Reach 40 Gb/s Transponder Electro-Absorption Modulator Driver	Released
P44R3	BW >65GHz, V p-p >3.0V 20 dB gain, Pdc 1.2W, <500 fs rms jitter, V-connectors	Electro-Absorption Modulator Driver Trans-Impedance Amplifier General Purpose Gain Block	Released
P423R3	BW >65GHz, V p-p >7.5V 30 dB gain, Pdc 3.5W, <500 fs rms jitter, V-connectors	Lithium Niobate Modulator Driver General Purpose PA	Released

New!



P44R3C EA Driver Amp
(0.94 x 0.62 x 0.26 in)



P423R3 EYE: <500fs rms jitter



P423R3 S-Parameters